

LA-DU3 / 3.2 (a) označím $V_1 = 1, V_2 = x, V_3 = x^2$.

V_i bude značit vektory ortonormální báze podprostoru $\text{LO}\{1, x, x^2\}$ získané G-S ortogonalizací.

$$U_1 = \frac{V_1}{\|V_1\|} = \frac{1}{\|1\|} = \frac{1}{\sqrt{\langle 1, 1 \rangle}} = \frac{1}{\sqrt{\int_1^5 1 \cdot 1 dx}} = \frac{1}{\sqrt{[x]_1^5}} = \frac{1}{2}$$

W_i bude značit ortogonální projekci V_i na $\text{LO}\{U_1, \dots, U_{i-1}\}$

$$W_2 = \langle U_1, V_2 \rangle U_1 \quad \langle U_1, V_2 \rangle = \int_1^5 \frac{1}{2} \cdot x dx = 6, \text{ takže } W_2 = 6 \cdot \frac{1}{2} = 3$$

$$V_2 - W_2 = x - 3 \quad U_2 = \frac{V_2 - W_2}{\|V_2 - W_2\|} = \frac{x - 3}{\|x - 3\|} = \frac{x - 3}{\sqrt{[x^2 - 6x + 9]_1^5}} = \frac{\sqrt{3}}{4} (x - 3)$$

$$\|x - 3\| = \sqrt{\langle x - 3, x - 3 \rangle} = \sqrt{\int_1^5 (x - 3)(x - 3) dx} = \frac{4}{\sqrt{3}} \quad \langle x - 3, x^2 \rangle \cdot \frac{\sqrt{3}}{4} \cdot \frac{\sqrt{3}}{4} (x - 3)$$

$$W_3 = \langle U_1, V_3 \rangle U_1 + \langle U_2, V_3 \rangle U_2 = \langle \frac{1}{2}, x^2 \rangle \frac{1}{2} + \langle \frac{\sqrt{3}}{4} (x - 3), x^2 \rangle \frac{\sqrt{3}}{4} (x - 3)$$

$$\langle \frac{1}{2}, x^2 \rangle = \int_1^5 \frac{x^2}{2} dx = \frac{62}{3} \quad \langle x - 3, x^2 \rangle = \int_1^5 x^3 - 3x^2 dx = 32$$

$$= \frac{62}{3} \cdot \frac{1}{2} + 32 \cdot \frac{3}{16} (x - 3) = \frac{32}{3} + 6x - 18 = 6x - \frac{23}{3}$$

$$V_3 - W_3 = x^2 - 6x + \frac{23}{3} \quad \|V_3 - W_3\| = \sqrt{\int_1^5 (x^2 - 6x + \frac{23}{3})^2 dx} = \sqrt{\frac{256}{45}} = \frac{16}{3\sqrt{5}}$$

$$U_3 = \frac{V_3 - W_3}{\|V_3 - W_3\|} = \frac{x^2 - 6x + \frac{23}{3}}{\frac{16}{3\sqrt{5}}} = \frac{3\sqrt{5}}{16} (x^2 - 6x + \frac{23}{3})$$

vektory $U_{1,2,3}$ tedy tvoří ortonormální bázi zadaneho podprostoru

(b) ortogonální projekci vektoru $\sin x$ na $\text{LO}\{U_1, U_2, U_3\}$ označím P a spočítám ji vstahem

$$P = \langle \frac{1}{2}, \sin x \rangle \frac{1}{2} + \langle \frac{\sqrt{3}}{4} (x - 3), \sin x \rangle \frac{\sqrt{3}}{4} (x - 3) + \langle \frac{3\sqrt{5}}{16} (x^2 - 6x + \frac{23}{3}), \sin x \rangle \frac{3\sqrt{5}}{16} (x^2 - 6x + \frac{23}{3})$$

$$\langle \frac{1}{2}, \sin x \rangle = \int_1^5 \frac{\sin x}{2} dx \approx 0,128, \quad \langle \frac{\sqrt{3}}{4} (x - 3), \sin x \rangle = \int_1^5 \frac{\sqrt{3}}{4} (x - 3) \sin x dx \approx -1,493$$

$$\langle \frac{3\sqrt{5}}{16} (x^2 - 6x + \frac{23}{3}), \sin x \rangle = \int_1^5 \frac{3\sqrt{5}}{16} (x^2 - 6x + \frac{23}{3}) \sin x dx \approx -0,125$$

$$\frac{\sqrt{3}}{4} \approx 0,433, \quad \frac{3\sqrt{5}}{16} \approx 0,419$$

$$P = 0,128 \cdot \frac{1}{2} - 1,493 \cdot 0,433 \cdot (x - 3) - 0,125 \cdot 0,419 (x^2 - 6x + 23) = 0,064 - 0,646x + 1,939 - 0,052x^2 + 0,374x - 1,204 = -0,052x^2 - 0,332x + 0,799$$

(c) $\|\sin x - P\| = \sqrt{\langle \sin x - P, \sin x - P \rangle} = \sqrt{\int_1^5 (\sin x - P)^2 dx} \approx 1,629$ označím Φ